# Homework 8

Due November 11, 2025.

## Problem 1.

We showed in class that $S_3$ and $B_3$ are topologically equivalent (with all points of the boundary of $B_3$ identified as only point), namely there is a 1–1 map between them which is continuous.

a) Now analyse the case $S_2$ and $B_2$.

b) Next analyse the case $S_1$ and $B_1$.

c) In which of these 3 cases is the sphere isomorphic to a group, and which group?

d) Which of these groups is a covering group, and which of them is not simply-connected?

## Problem 2.

We proved that $SU(2)/Z_2 \simeq SO(3)$ by considering the equation

$$u^{-1}x^k\sigma_k u = x'^k\sigma_k = R^i{}_j x^j\sigma_i\,, \qquad u = e^{-\frac{i}{2}\omega^k\sigma_k}\,. \tag{1}$$

a) Prove that the left-hand side is linear in $\sigma_l$ (expand $e^{-\frac{i}{2}\omega^k\sigma_k}$) and determine $R^i{}_j$.

b) Prove that $R$ is a rotation (orthogonal with $\det R = 1$) and does not rotate the axis of rotation.
*Hint:* Consider $\operatorname{tr} x\cdot\sigma x\cdot\sigma$ and $\operatorname{tr} x'\cdot\sigma x'\cdot\sigma$.

c) Consider the homomorphism $\phi(u) = R$ in a). What is the kernel?

## Problem 3. The Lorentz group $L = O(3,1)$.

The group $O(3,1)$ preserves $x^\mu\eta_{\mu\nu}x^\nu$ with $\eta_{\mu\nu} = (-1,+1,+1,+1)$, hence $x'^\mu = L^\mu{}_\nu x^\nu$ with $L^T\eta L = \eta$.

a) Show that $\det L = \pm 1$.

b) Show that $L_{00} = \eta_{00}L^0{}_0$ is either $\geq 1$ or $\leq -1$.

c) So $O(3,1)$ contains 4 disconnected components. Write

$$O(3,1) = SO(3,1)^\uparrow \cup \Lambda_P SO(3,1)^\uparrow \cup \Lambda_T SO(3,1)^\uparrow \cup \Lambda_{PT} SO(3,1)^\uparrow\,. \tag{2}$$

Find diagonal matrices for $\Lambda_P$, $\Lambda_T$ and $\Lambda_{PT}$.

d) Consider the homomorphism $\phi : L \to (\det L, \text{sign } L_{00})$. Which group $V$ do the four group elements $(\det \Lambda, \Lambda_{00}) = (\pm 1, \pm 1)$ form?

e) Now show that $SO(3,1)^{\uparrow}$ is the kernel of the homomorphism $O(3,1)/V$. Is $SO(3,1)^{\uparrow}$ a normal subgroup of $O(3,1)$?

## Problem 4.

We want to prove that $Sl(2,\mathbb{C})$ is the covering group of $SO(3,1)^{\uparrow}$. Consider the expression $Ax^{\mu}\sigma_{\mu}A^{\dagger}$ with $A$ in $Sl(2,\mathbb{C})$ and $\sigma_{\mu} = (-I, \sigma_1, \sigma_2, \sigma_3)$. (Note $A^{\dagger}$, not $A^{-1}$ this time.)

a) Show that this expression is equal to $x'^{\mu}\sigma_{\mu}$, where $x'^{\mu} = L^{\mu}{}_{\nu}x^{\nu}$ with **real** $L^{\mu}{}_{\nu}$.
*Hint:* Can one always write a complex $2 \times 2$ matrix as $x'^{\mu}\sigma_{\mu}$?

b) Show that $\det x^{\mu}\sigma_{\mu} = \det x'^{\mu}\sigma_{\mu}$, and then show that $L^{\mu}{}_{\nu} \in O(3,1)$.

c) Now show that $L^{\mu}{}_{\nu} \in SO(3,1)^{\uparrow}$. We denote these group elements by $\hat{L}^{\mu}{}_{\nu}$.

d) Show that the map $\phi(A) = \hat{L}$ is a homomorphism. What is the kernel of $\phi$?

e) Show that $Sl(2,\mathbb{C})/Z_2 \simeq SO(3,1)^{\uparrow}$.

## Bonus Problem.

Show that $Sl(2,\mathbb{C})$ is simply connected.

a) First assume that $A \in Sl(2,\mathbb{C})$ has the polar decomposition $A = HU$ where $U \in SU(2)$ and $H$ is a $2 \times 2$ matrix which is hermitian, has $\det H = 1$, and $\operatorname{tr} H > 0$. Show that the manifold of $H$ is a connected component of a hyperboloid. Then use this to show that $Sl(2,\mathbb{C})$ is simply connected.

b) Show that the matrices $H$ do not form a group (if group multiplication is matrix multiplication).

c) Now prove the polar decomposition $A = HU$.
*Hint:* First diagonalize $AA^{\dagger}$ by a unitary matrix $V$ such that $VAA^{\dagger}V^{-1} = D^2$. Let $D$ be the positive square root of $D^2$, so $\operatorname{tr} D > 0$. Then define $H = V^{-1}DV$.

d) Show that $U = H^{-1}A$ lies in $SU(2)$.

e) We showed above that $Sl(2,\mathbb{C})$ is simply connected. Is $Gl(2,\mathbb{C})$ also simply connected?